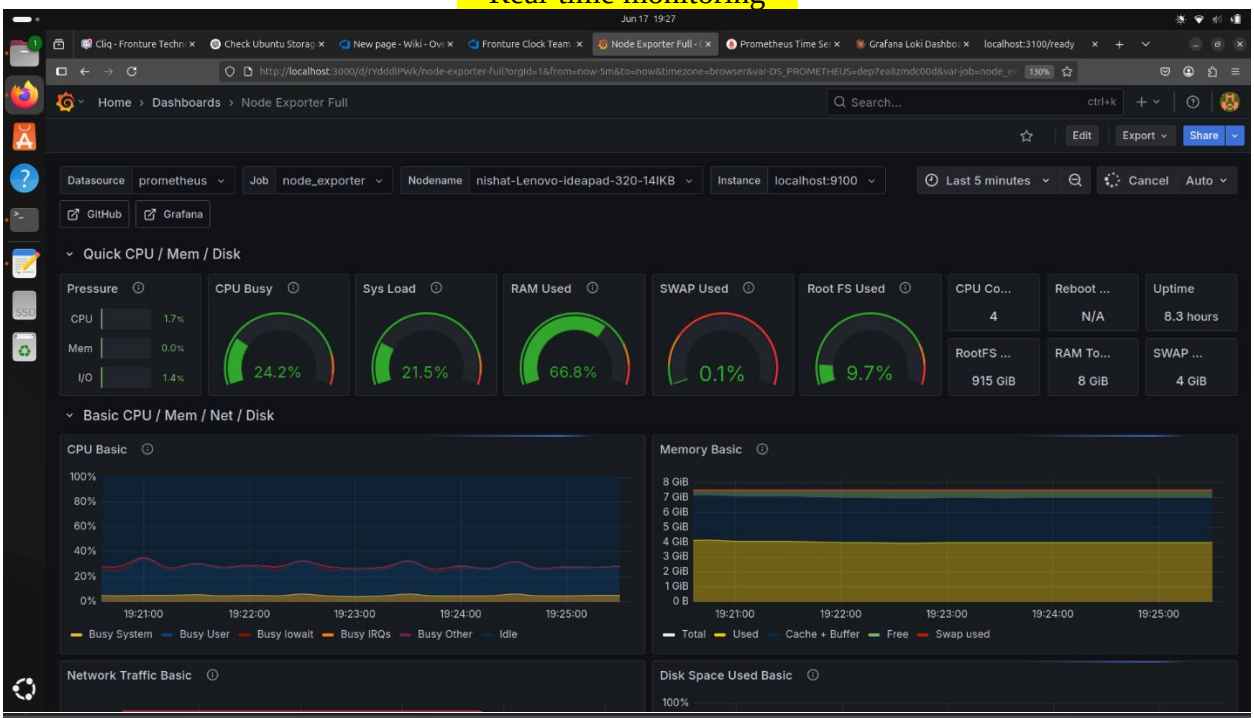
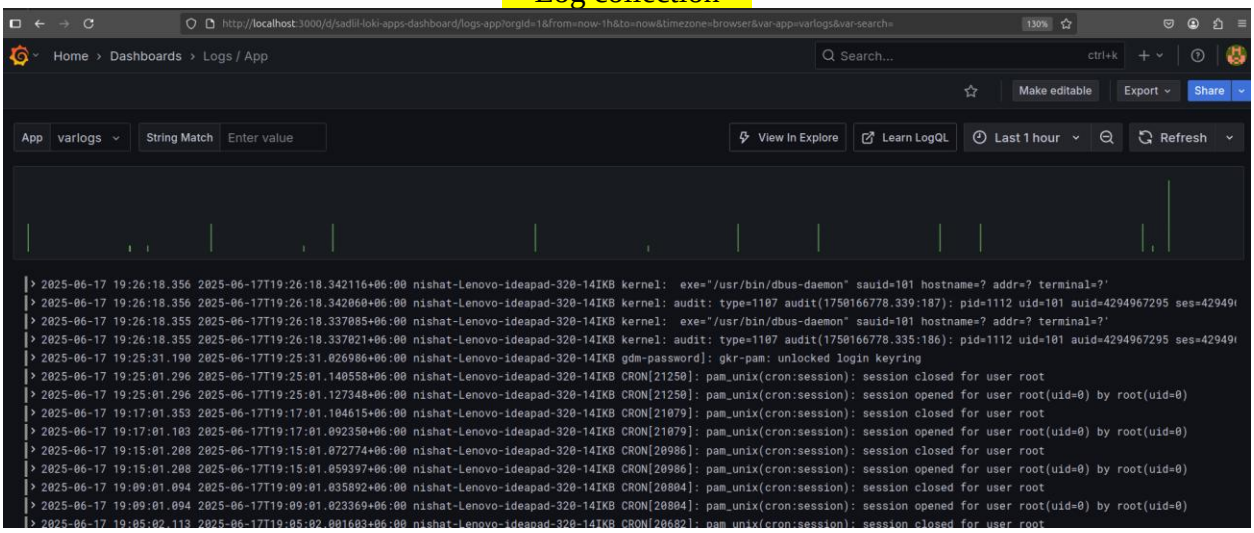
Linux vm performance metric using Prometheus, Loki, and Grafana

Documented by: NISHAT AFLA

Real-time monitoring



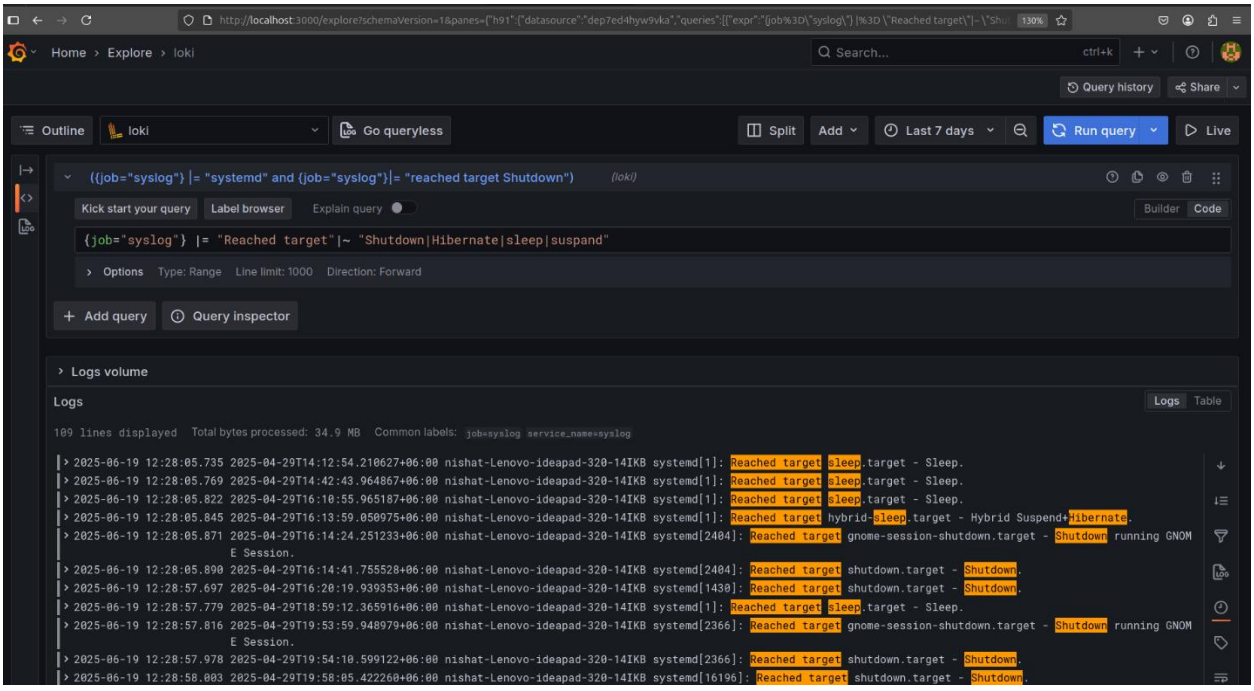
Log collection



##

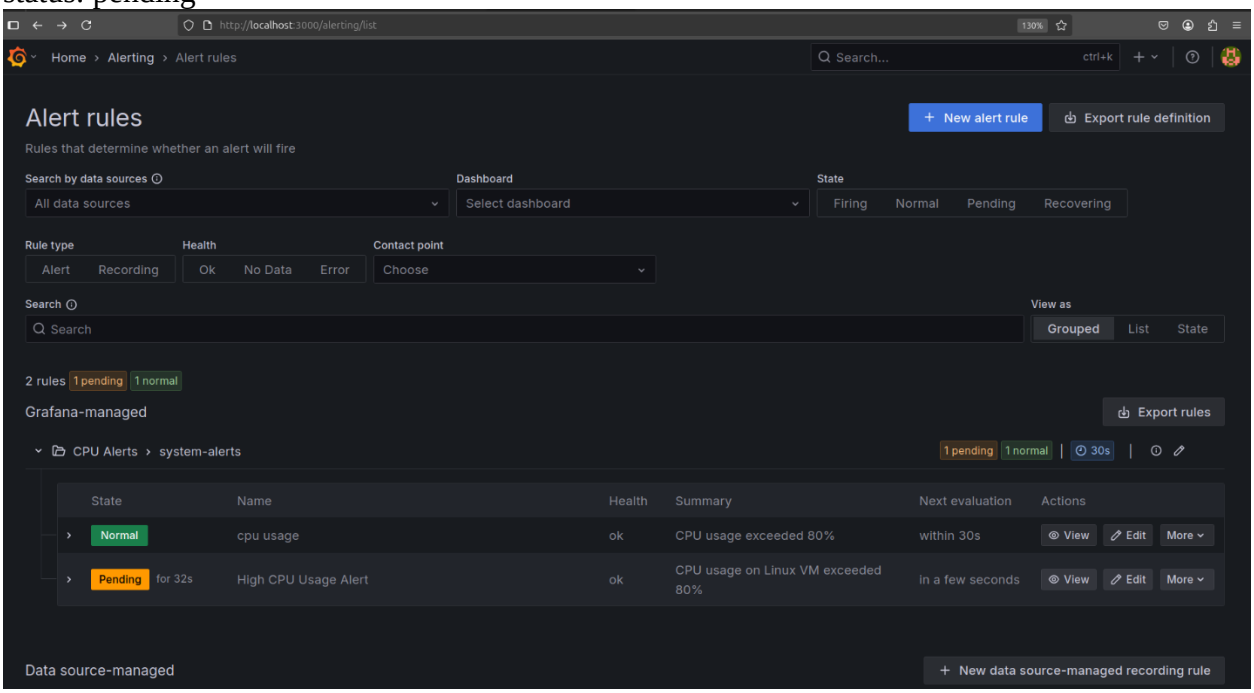
Event logs (last 7 days vm on/off events) visualization using grafana:

>**LogQL Query**: {job="syslog"} |= "Reached target"|~ "Shutdown|Hibernate|sleep|suspand"

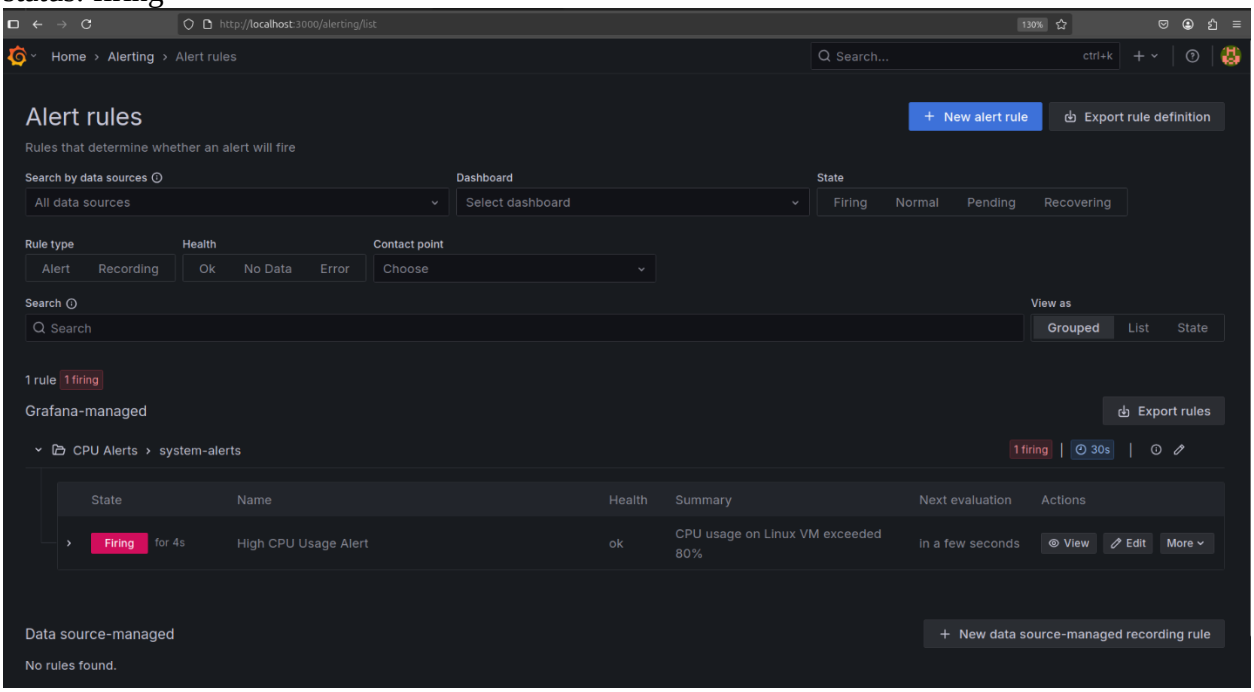


Alerting set up if cpu usage exceed 80%

status: pending



status: firing



SETUP

Install Node Exporter (System Metrics Collector)

Node Exporter collects CPU, RAM, disk, and network metrics from your Linux VM and exposes them to Prometheus.

Download Node Exporter
`cd /opt`

`sudo wget https://github.com/prometheus/node_exporter/releases/download/$latest_version/node_exporter-${latest_version#v}.linux-amd64.tar.gz
`

Extract and Move Binary

`sudo tar -xvf node_exporter-*.tar.gz -C /opt`
Move the binary to `/usr/local/bin`:
`sudo mv /opt/node_exporter-*.linux-amd64/node_exporter /usr/local/bin/`
Check the binary works:
`/usr/local/bin/node_exporter --version`

create a systemd service so node_exporter starts automatically and runs in the background.

Create the service file:
`sudo nano /etc/systemd/system/node_exporter.service`

Paste this:

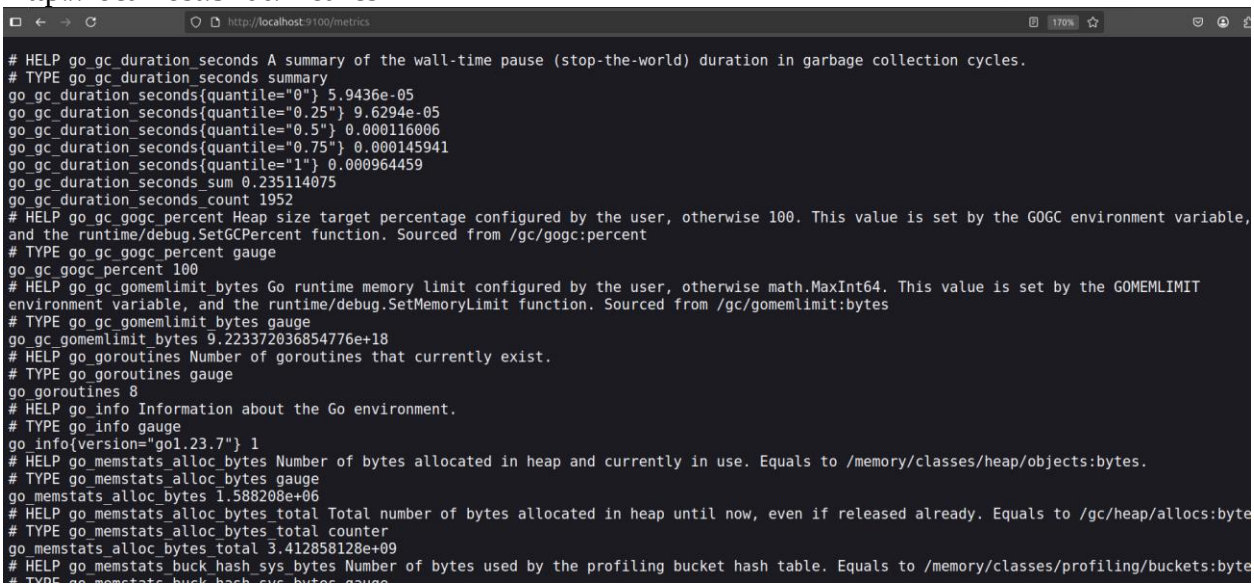
>[Unit]
Description=Node Exporter
After=network.target
[Service]
User=nobody {replace `User=nobody` with your actual Ubuntu username if different.}
ExecStart=/usr/local/bin/node_exporter
[Install]
WantedBy=default.target

Start and Enable the Service

sudo systemctl daemon-reexec
sudo systemctl daemon-reload
sudo systemctl enable node_exporter
sudo systemctl start node_exporter

Verify it's Running

http://localhost:9100/metrics



Install and configure Prometheus to scrape Node Exporter metrics.**
Install Prometheus

`cd /opt`

Download the latest valid Prometheus version
`sudo wget https://github.com/prometheus/prometheus/releases/download/v2.52.0/prometheus-2.52.0.linux-amd64.tar.gz`

Extract it
`sudo tar xvf prometheus-2.52.0.linux-amd64.tar.gz`
Go into the directory
`cd prometheus-2.52.0.linux-amd64`

_`sudo nano prometheus.yml`

paste the content:
>global:
  scrape_interval: 15s
scrape_configs:
> - job_name: "prometheus"

```
  static_configs:
    - targets: ["localhost:9090"]

> - job_name: "node_exporter"
  static_configs:
    - targets: ["localhost:9100"]
```

Start Prometheus

```
>sudo chown -R $ USER:$ USER /opt/prometheus-2.52.0.linux-amd64/data
>./prometheus --config.file=prometheus.yml
```

You should see Prometheus starting logs.

Access Prometheus UI

```
>sudo bash -c "./prometheus --config.file=prometheus.yml > /opt/prometheus.log 2>&1 &"
```

Open your browser and go to:

👉 http://localhost:9090

Install Loki (for logs collection)

Loki will collect and store logs — Promtail will send system logs to it.

Download Loki

```
cd /opt
latest_loki=$ (curl -s https://api.github.com/repos/grafana/loki/releases/latest | grep tag_name |
cut -d "" -f4)
wget https://github.com/grafana/loki/releases/download/$latest_loki/loki-linux-amd64.zip
```

Unzip Loki

```
sudo apt install unzip -y
unzip loki-linux-amd64.zip
chmod +x loki-linux-amd64
```

Create Loki Config File

Create the file:

```
nano loki-config.yaml
```

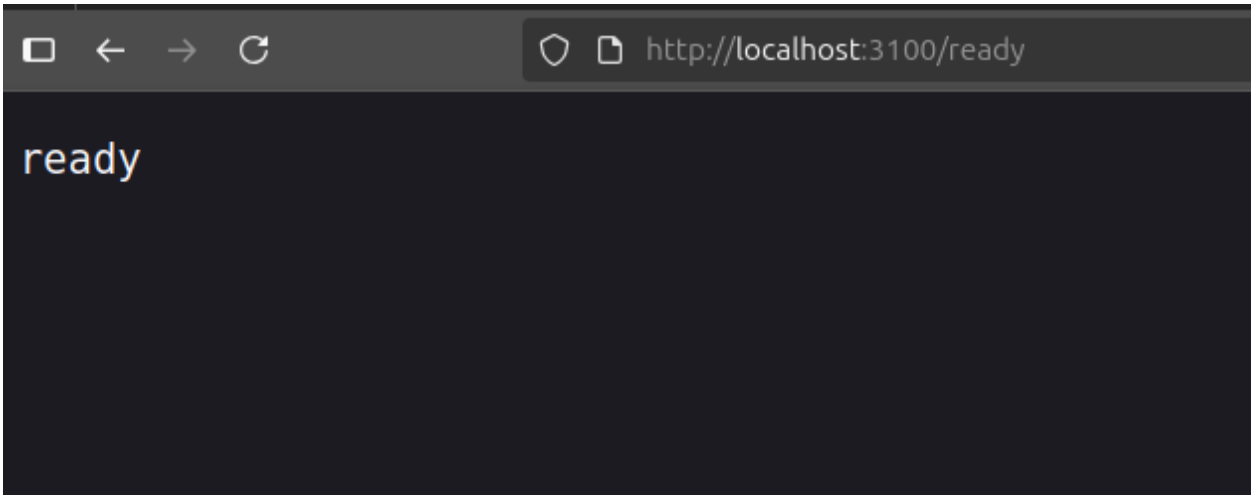
Paste the following content:

```
auth_enabled: false

server:
  http_listen_port: 3100
```

```
  log_level: info

ingester:
  lifecycler:
    ring:
      kvstore:
        store: inmemory
      replication_factor: 1
  chunk_idle_period: 5m
  max_chunk_age: 1h
  wal:
    dir: /tmp/wal

schema_config:
  configs:
    - from: "2022-01-01"
      store: boltdb
      object_store: filesystem
      schema: v11
      index:
        prefix: index_
        period: 24h

storage_config:
  boltdb:
    directory: /tmp/loki/index
  filesystem:
    directory: /tmp/loki/chunks

limits_config:
  reject_old_samples: true
  reject_old_samples_max_age: 168h
  allow_structured_metadata: false

table_manager:
  retention_deletes_enabled: true
  retention_period: 168h
```

Start Loki in background

```
>sudo bash -c "./loki-linux-amd64 -config.file=loki-config.yaml > loki.log 2>&1 &"
```



Promtail Setup
cd /opt
sudo wget https://github.com/grafana/loki/releases/download/v3.5.1/promtail-linux-amd64.zip

Then unzip it (also with sudo):

sudo unzip promtail-linux-amd64.zip

And make it executable:

sudo chmod +x promtail-linux-amd64

Create a file named `promtail-config.yaml` inside `/opt`:
>sudo nano /opt/promtail-config.yaml

```
server:
  http_listen_port: 9080
  grpc_listen_port: 0

positions:
  filename: /tmp/positions.yaml

clients:
  - url: http://localhost:3100/loki/api/v1/push

scrape_configs:
  - job_name: system-logs
    static_configs:
      - targets:
          - localhost
        labels:
          job: varlogs
          __path__: /var/log/*.log
```

This will collect logs from files like `/var/log/syslog`, `/var/log/auth.log`, etc., and send them to Loki at `localhost:3100`.

run promptail in background:
sudo bash -c "./promtail-linux-amd64 -config.file=promtail-config.yaml > promtail.log 2>&1 &"

Grafana:
Manually add the GPG key for Grafana:

`sudo mkdir -p /etc/apt/keyrings
curl -fsSL https://packages.grafana.com/gpg.key | gpg --dearmor | sudo tee /etc/apt/keyrings/grafana.gpg > /dev/null`

Update your Grafana source to use the keyring:

`echo "deb [signed-by=/etc/apt/keyrings/grafana.gpg] https://packages.grafana.com/oss/deb stable main" | \
sudo tee /etc/apt/sources.list.d/grafana.list`

Update package list and install Grafana:

`sudo apt-get update`
`sudo apt-get install grafana`

Start and enable Grafana service:

`sudo systemctl start grafana-server`
`sudo systemctl enable grafana-server`

Access Grafana:
Open your browser and go to: http://localhost:3000
Default login cred:

Username: admin

Password: admin

Now it’s time to **verify each component and connect them inside **Grafana****

Visit this in your browser:
http://localhost:9090 → Prometheus web UI should open.
* Click **“Status” > “Targets”**
  * You should see a **target like `localhost:9100`** (Node Exporter)

  * **Health = "UP"** means Prometheus is pulling metrics from Node Exporter

in Grafana:
1. **Go to (Settings) > Data Sources**

Add datasource as prometheus:
* **Prometheus**
  * URL: `http://localhost:9090`

Loki
  * URL: `http://localhost:3100`

Click “Save & Test” on both. If successful = ☑
Import Dashboards

Now you can import pre-built dashboards:
1. Go to Import**

2. Use these Dashboard IDs:
   * **Node Exporter** → ID: `1860`

   * **Loki logs dashboard** → ID: `13639`

3. Select your data source (Prometheus for Node Exporter, Loki for logs)

Done you’ll now see live CPU, memory, disk usage + logs!